

Informative hypotheses evaluation

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ANOVA Example: Comparisons of 3 Means

Example Palmer and Gough (P&G)

Palmer and Gough (2007) examined the difference in the attribution of importance of defective education as an explanation for criminal behavior between three types of “offenders”:
(1) non-offenders, (2) property offenders, and (3) person offenders.

 $(n_i = 20, 20, 31)$

Hypotheses of interest

Example Palmer and Gough (P&G)

Palmer and Gough (2007) expect that

- non-offenders (1) attribute more importance to defective education for explaining crime than the other two offenders (2 & 3):
i.e., $\mu_1 > \mu_2$ and $\mu_1 > \mu_3$,
- property offenders (2) attribute more importance to defective education than person offenders (3):
i.e., $\mu_2 > \mu_3$.

This leads to the theory-based hypothesis:

$$H_1 : \mu_1 > \mu_2 > \mu_3.$$

Note that ‘ $<$ ’ denotes “smaller than” and ‘ $>$ ’ denotes “larger than”.

Testing the null hypothesis

Example 3 means

Test H_0 with ANOVA F test:

$$H_0 : \quad \mu_1 = \mu_2 = \mu_3,$$

$$H_a : \quad \text{not } H_0.$$

Then, reject or not-reject ('accept') H_0 .

But, often not interested in H_0 !

Cannot say anything about $H_1 : \mu_1 > \mu_2 > \mu_3$.

NHST

Example Palmer and Gough

```
PandG_data <- read.table("Data_PalmerAndGough.txt",
                          header=TRUE)
PandG_data$group <- factor(PandG_data$group)
pairwise.t.test(PandG_data$Importance, PandG_data$group,
                p.adj = 'bonferroni')

##
## Pairwise comparisons using t tests with pooled SD
##
## data: PandG_data$Importance and PandG_data$group
##
##      1      2
## 2 0.191 -
## 3 0.011 1.000
##
## P value adjustment method: bonferroni
```


Exploration and Confirmation

Exploration (like in post hoc tests and possibly AIC)

Evaluate all possible pairs/subsets of means whether significant different (“,”) or not (“=”).

For example, when $k = 3$:

$$H_{0E} : \mu_1 = \mu_2 = \mu_3$$

$$H_{1E} : \mu_1 = \mu_2, \mu_3$$

$$H_{2E} : \mu_1, \mu_2 = \mu_3$$

$$H_{3E} : \mu_1 = \mu_3, \mu_2$$

$$H_a : \mu_1, \mu_2, \mu_3$$

When $k = 5$, there are even 52 hypotheses.

Confirmation

Limited set: Evaluate only prespecified hypotheses including order restrictions ($<$, $>$, but also $=$).

Confirmatory methods

Most researchers are able to specify “order-restricted” / “informative” / “theory-based” hypotheses, like $H_1 : \mu_1 > \mu_2 > \mu_3$.
Use prior knowledge and/or expertise in hypothesis.

Note: 'model' refers to hypothesis.

GORIC

Example Palmer and Gough, that is, 3 means

$$H_0 : \mu_1 = \mu_2 = \mu_3,$$

$$H_1 : \mu_1 > \mu_2 > \mu_3,$$

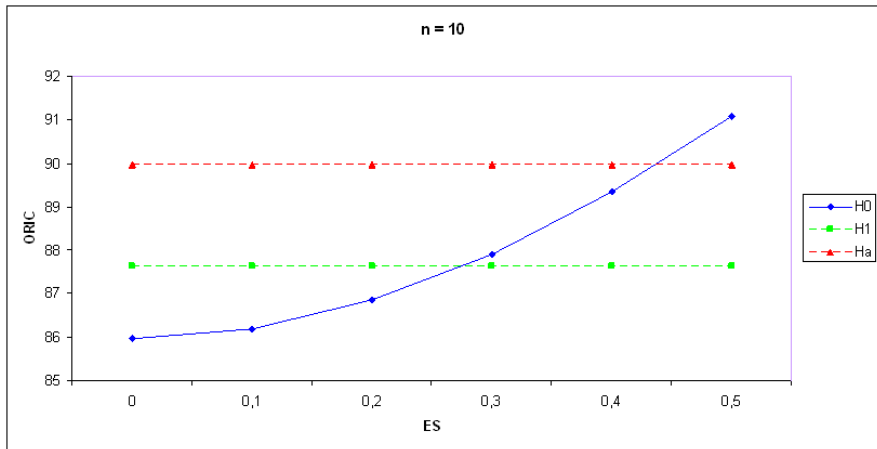
$$H_u : \mu_1, \mu_2, \mu_3.$$

GORIC – will be explained later

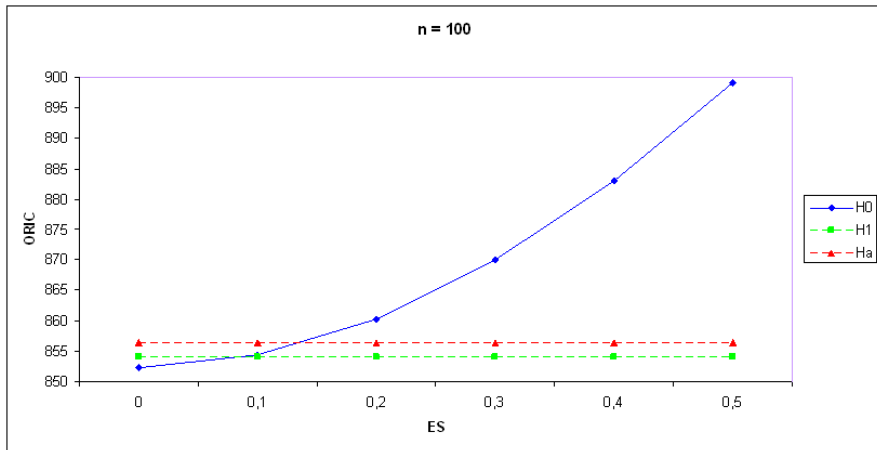
Model	Fit	Complexity	GORIC	GORIC weights
H_0	-196.36	2.00	396.71	0.02
H_1	-191.89	2.81	389.41	0.75
H_u	-191.89	4.00	391.79	0.23

Confirmatory methods (e.g., GORIC) have more “power” than their exploratory counterparts (e.g., AIC; cf. one- vs two-sided testing).

Confirmation more power: 1 data set. GORIC values for 3 groups, effect size ES , and $n = 10$ observations per group



Confirmation more power: 1 data set. GORIC values for 3 groups, effect size ES , and $n = 100$ observations per group

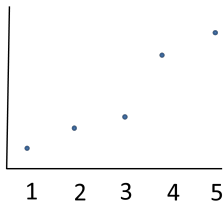


- Kuiper, R. M., and Hoijtink, H. (2010). Comparisons of Means Using Exploratory and Confirmatory Approaches. *Psychological Methods*, 15(1), 69–86.
- Kuiper, R. M., Klugkist, I., and Hoijtink, H. (2010). A Fortran 90 Program for Confirmatory Analysis of Variance. *Journal of Statistical Software*, 34(8), 1–31.
- Kuiper, R.M., Hoijtink, H. and Silvapulle, M.J. (2011). An Akaike type information criterion for model selection under inequality constraints. *Biometrika*, 98, 495-501. (GORIC)
- Kuiper, R.M., Nederhof, T., and Klugkist, I. (2015). Properties of hypothesis testing techniques and (Bayesian) model selection for exploration-based and theory-based (order-restricted) hypotheses. *British Journal of Mathematical and Statistical Psychology*, 68(2), 220 – 245.
- Altınışık, Y., Van Lissa, C. J., Hoijtink, H., Oldehinkel, A. J., and Kuiper, R. M. (2021). Evaluation of inequality constrained hypotheses using a generalization of the AIC. *Psychological Methods*, 26(5), 599–621.
<https://doi.org/10.1037/met0000406> (GORICA)

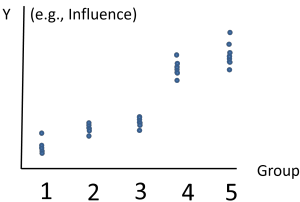
Intermezzo: Balance Fit and Complexity (1/6)

Example 5 means

Data for 5 groups – Oversimplified representation

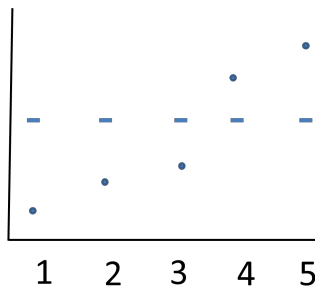


Data for 5 groups – Better representation (scatter of data points)



Intermezzo: Balance Fit and Complexity (2/6)

Example 5 means

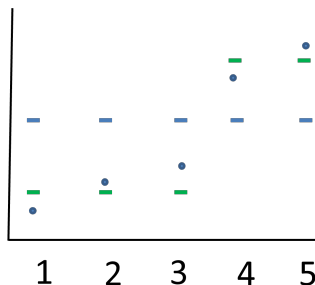


$$\mu_1 = \mu_2 = \mu_3 = \mu_4 = \mu_5 = \mu$$

parameters: 1 mean parameter (i.e., μ)
Fit: Bad fit

Intermezzo: Balance Fit and Complexity (3/6)

Example 5 means



$$\mu_1 = \mu_2 = \mu_3 = \mu_{\text{low}}$$

$$\mu_4 = \mu_5 = \mu_{\text{high}}$$

parameters:

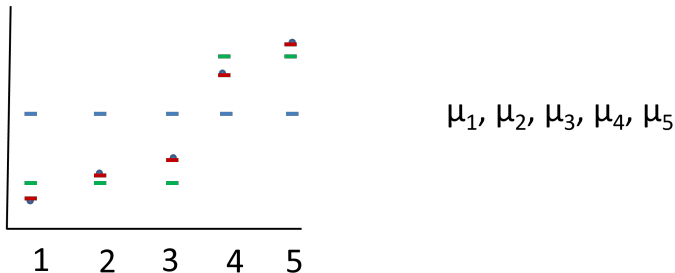
2 mean parameters (i.e., μ_{low} and μ_{high})

Fit:

Better than with one parameter, even good fit.

Intermezzo: Balance Fit and Complexity (4/6)

Example 5 means



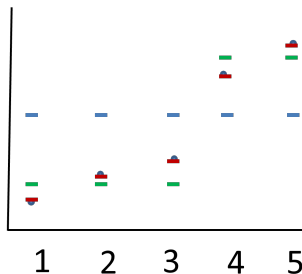
parameters: 5 mean parameter (complex)

Fit: Better than with two parameters, even best possible fit.

Thus: Best fit (= highest likelihood), but also most complex (= highest penalty).

Intermezzo: Balance Fit and Complexity (5/6)

Example 5 means



$$\mu_1 = \mu_2 = \mu_3 = \mu_4 = \mu_5 = \mu$$

$$\mu_1 = \mu_2 = \mu_3 = \mu_{\text{low}} \text{ \& }$$

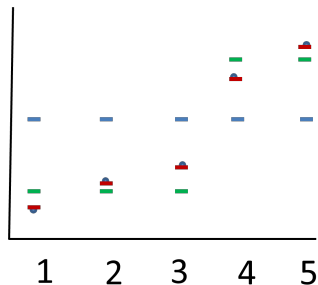
$$\mu_4 = \mu_5 = \mu_{\text{high}}$$

$$\mu_1, \mu_2, \mu_3, \mu_4, \mu_5$$

In this example: **Two means** may be best trade-off between fit & complexity.

Intermezzo: Balance Fit and Complexity (6b/6)

Order-restrictions: From “Example 5 means” to ‘Example 2 means’



In theory-based hypotheses, we also incorporate **order-restrictions** (e.g., $\mu_1 < \mu_2$). Then, helpful to look at likelihood using **contour plots** (not scatter plot, as done here).

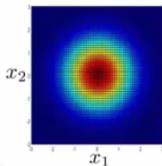
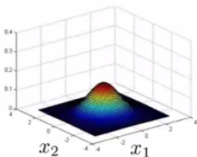
For ease, I will next use **2 means** (instead of 5).

Intermezzo: Contour plot (2/2)

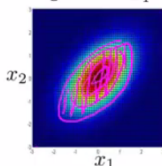
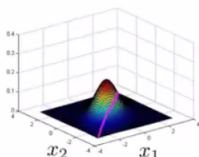
Example 2 means

Multivariate Gaussian (Normal) examples

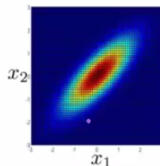
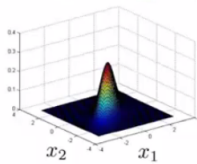
$$\mu = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \Sigma = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$



$$\mu = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \Sigma = \begin{bmatrix} 1 & 0.5 \\ 0.5 & 1 \end{bmatrix}$$



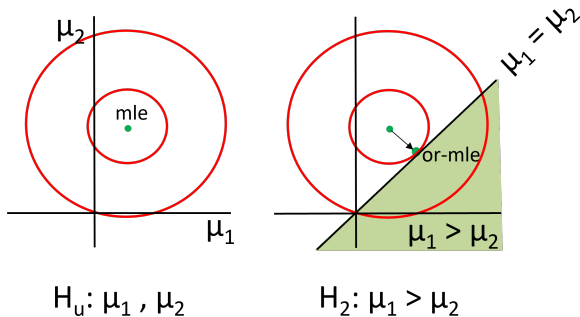
$$\mu = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \Sigma = \begin{bmatrix} 1 & 0.8 \\ 0.8 & 1 \end{bmatrix}$$



Andreas

Idea fit: order-restricted maximum likelihood (or-ml)

Example 2 means (with or-mle $\neq$ mle)



mle not in H_2 : Find highest likelihood in allowable (= green) space.
 The resulting estimated mean $\hat{\mu} = (\hat{\mu}_1, \hat{\mu}_2)$ is referred to as or-mle.

Idea fit

Example Palmer and Gough, that is, 3 means

$$H_0 : \mu_1 = \mu_2 = \mu_3,$$

$$H_1 : \mu_1 > \mu_2 > \mu_3,$$

$$H_2 : \mu_1 > \mu_2 < \mu_3,$$

$$H_3 : \mu_1 < \mu_2 < \mu_3,$$

$$H_u : \mu_1, \mu_2, \mu_3.$$

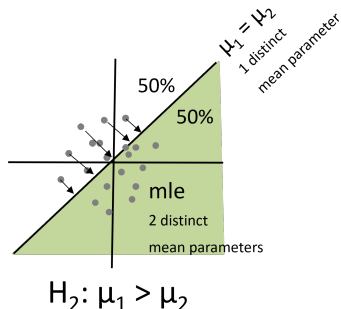
```
## group1 group2 group3
## 11.95  9.75  8.77
```

GORIC

Model	Fit	Complexity	GORIC	GORIC weights
H_0	-196.36	2.00	396.71	0.01
H_1	-191.89	2.81	389.41	0.56
H_2	-192.34	3.19	391.05	0.25
H_3	-196.36	2.81	398.34	0.01
H_u	-191.89	4.00	391.79	0.17

Idea complexity: penalty (PT)

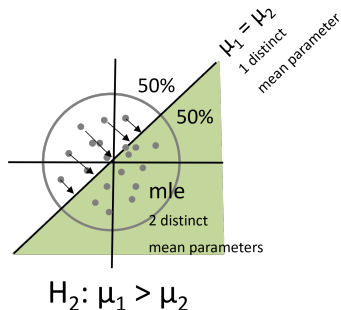
Example 2 means



Note: sampling from distribution with $\mu_1 = \mu_2 = 0$.
 Complexity H_2 is $PT_2 = 1 + 0.5 \times 1 + 0.5 \times 2 = 2.5$.
 PT = number of expected distinct/free parameters.

Idea complexity: penalty (PT)

Example 2 means: Contour distribution



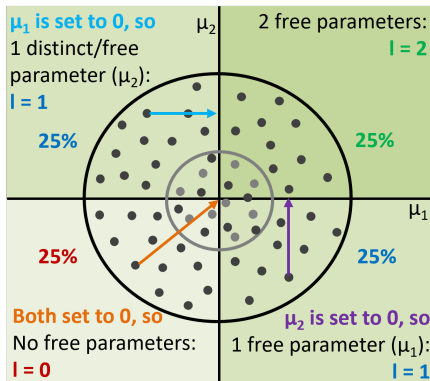
Sampling from distribution with $\mu_1 = \mu_2 = 0$. (Note: not likelihood!)

Contour denotes distribution for a variance value.

Choice variance does not matter, see next slide.

Extra: Levels and level probabilities

Example 2 means



$H_4: \mu_1 > 0, \mu_2 > 0$

Level (l): number of distinct/free parameters.

Level probability (LP_l): probability that there are l levels.

$PT_4 = 1 + 0.25 \times 0 + (0.25 + 0.25) \times 1 + 0.25 \times 2 = 2.$

Idea complexity

Example Palmer and Gough, that is, 3 means

$$H_0 : \mu_1 = \mu_2 = \mu_3,$$

$$H_1 : \mu_1 > \mu_2 > \mu_3,$$

$$H_2 : \mu_1 > \mu_2 < \mu_3,$$

$$H_3 : \mu_1 < \mu_2 < \mu_3,$$

$$H_u : \mu_1, \mu_2, \mu_3.$$

GORIC

Model	Fit	Complexity	GORIC	GORIC weights
H_0	-196.36	2.00	396.71	0.01
H_1	-191.89	2.81	389.41	0.56
H_2	-192.34	3.19	391.05	0.25
H_3	-196.36	2.81	398.34	0.01
H_u	-191.89	4.00	391.79	0.17

GORIC

Example Palmer and Gough

- $H_0 :$ $\mu_1 = \mu_2 = \mu_3$ (include only when of interest)
 $H_1 :$ $\mu_1 > \mu_2 > \mu_3$
 $H_2 :$ $\mu_2 > \mu_1 > \mu_3$ (if competing theory/hypothesis)
 $H_u :$ μ_1, μ_2, μ_3 .

GORIC

Model	Fit	Complexity	GORIC
H_0	-196.36	2.00	396.71
H_1	-191.89	2.81	389.41
H_2	-193.70	2.81	393.03
H_u	-191.89	4.00	391.79

Hands-on/Demo (1a): GORIC

- Go to <https://github.com/rebeccakuiper/Tutorials>:
 - Click on green button called Code.
 - Download zip (last option in list).
 - Unzip it on your machine (that folder is now your working dir.).
- Start Rstudio. Optional: make project.
- Open 'Tutorial_GORIC_restriktor_ANOVA.html', 'PalmerAndGough_and_Lucas.R', and/or '**Hands-on_1_GORIC_Unc_ANOVA_restriktor.R**' ('Hands-on files').
- Install packages and load them.
- Read and inspect data.
Use **Data_PalmerAndGough.txt** and/or Data_Lucas.txt.
- Run model (lm()).
- Specify hypotheses (make up your own).
Note: Use names used in the model.
- Run goric().
- Inspect and interpret output.
Note: 'GORIC weights' will be explained next.



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- If needed: Start Rstudio again (optional: make project) and then also load packages again.
- Read and inspect data.
- Run model (`lm()`).
- Specify hypotheses (make up your own).
Note: Use names used in the model.
- Run `goric()`.
- Inspect and interpret output: Focus on GORIC weights.

GORIC output and interpretation

Palmer & Gough

$$H_1 : \mu_1 > \mu_2 > \mu_3$$

$$H_2 : \mu_2 > \mu_1 > \mu_3$$

$$H_u : \mu_1, \mu_2, \mu_3$$

GORIC

Model	Fit	Complexity	GORIC	GORIC weights
H_1	-191.89	2.81	389.41	0.68
H_2	-193.70	2.81	393.03	0.11
H_u	-191.89	4.00	391.79	0.21

H_1 and H_2 are not weak (nl, better than H_u - will be explained in a bit).

H_1 is $0.68 / 0.11 \approx 6.11$ times more supported than H_2 .

(Two informative) hypotheses of interest

$H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4 = \mu_5$ (include only when of interest)

$H_1 : \mu_5 = \mu_3 > \{\mu_1, \mu_4\} > \mu_2$

$H_2 : \mu_3 > \mu_1 > \mu_4 = \mu_5 > \mu_2$

$H_u : \mu_1, \mu_2, \mu_3, \mu_4, \mu_5$ (included as failsafe)

Descriptive statistics of Lucas' Data

Group	Mean Influence	SD	n
1 (male, random)	2.33	1.86	30
2 (female, random)	1.33	1.15	30
3 (male, selected)	3.20	1.79	30
4 (female, selected)	2.23	1.45	30
5 (female, selected+)	3.23	1.50	30

GORIC weights (w_m)

Lucas

GORIC

Model	Fit	Complexity	GORIC	GORIC weights
H_1	-278.05	3.46	563.02	0.90
H_2	-281.76	3.14	569.79	0.03
H_u	-278.05	6.00	568.10	0.07

H_1 is $0.9 / 0.03 \approx 29.59$ times more supported than H_2 .
Thus, there is quite some evidence that H_1 is the best of this set.

Note:

H_1 is $0.9 / 0.07 > 1$ times more supported than H_u .
Hence, H_1 not weak and thus comparable to the other
(weak and non-weak) informative hypotheses (as explained in a bit).

The failsafe hypothesis will be preferred over the informative hypotheses in the set, if the sample size is large enough.

E.g.: $w_1 = .08$, $w_2 = .02$, and $w_u = .90$.

Use of H_u

Palmer & Gough

$$\begin{aligned}
 H_1 : & \quad \mu_1 > \mu_2 > \mu_3, \\
 H_2 : & \quad \mu_2 > \mu_1 > \mu_3 \\
 H_u : & \quad \mu_1, \mu_2, \mu_3.
 \end{aligned}$$

GORIC

Model	Fit	Complexity	GORIC	GORIC weights
H_1	-191.89	2.81	389.41	0.68
H_2	-193.70	2.81	393.03	0.11
H_u	-191.89	4.00	391.79	0.21

If at least one informative hypothesis not weak (i.e., $w_m > w_u$ or $w_m/w_u > 1$), then compare informative hypotheses.

Hence: H_u is only a failsafe, not another hypothesis of interest.

Use of H_u

Palmer & Gough

$$H_1 : \mu_1 > \mu_2 > \mu_3,$$

$$H_2 : \mu_2 > \mu_1 > \mu_3$$

$$H_u : \mu_1, \mu_2, \mu_3.$$

GORIC

Model	Fit	Complexity	GORIC	GORIC weights	without Hu
H_1	-191.89	2.81	389.41	0.68	0.86
H_2	-193.70	2.81	393.03	0.11	0.14
H_u	-191.89	4.00	391.79	0.21	NA

H_1 is not weak and is $0.68 / 0.11 = 0.86 / 0.14 \approx 6.11$ times more supported than competing hypothesis H_2 .

H_1 vs H_u

Palmer & Gough

What if only one informative hypothesis:

$$H_1 : \mu_1 > \mu_2 > \mu_3,$$

$$H_u : \mu_1, \mu_2, \mu_3.$$

GORIC

Model	Fit	Complexity	GORIC	GORIC weights
H_1	-191.89	2.81	389.41	0.77
H_u	-191.89	4.00	391.79	0.23

H_1 is $0.77 / 0.23 \approx 3.27$ times more supported than H_u .

BUT: H_u includes H_1 .

So, support for H_u contains support for H_1 .

H_1 vs H_u

Palmer & Gough

What if only one informative hypothesis:

$$H_1 : \mu_1 > \mu_2 > \mu_3,$$

$$H_u : \mu_1, \mu_2, \mu_3.$$

GORIC

Model	Fit	Complexity	GORIC	GORIC weights
H_1	-191.89	2.81	389.41	0.77
H_u	-191.89	4.00	391.79	0.23

H_1 is $0.77 / 0.23 \approx 3.27$ times more supported than H_u .

BUT: H_u includes H_1 .

So, support for H_u contains support for H_1 .

H_1 vs H_u ctd.

Palmer & Gough

GORIC

Model	Fit	Complexity	GORIC	GORIC weights
H_1	-191.89	2.81	389.41	0.77
H_u	-191.89	4.00	391.79	0.23

Unconstrained is always true, so will always receive support.
Support for H_{μ} contains support for H_1 (if any).

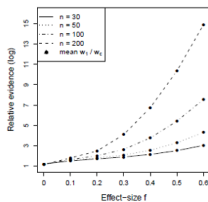
Now: Fit H_1 and H_u the same;
thus, GORIC weights are based on only complexity.

Consequently, $0.77 / 0.23 \approx 3.27$ is an upper bound.

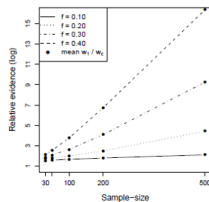
Alternative failsafe: Complement of H_m

Based on simulation

vs complement

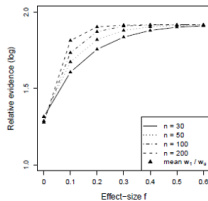


(a)

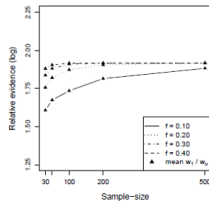


(b)

vs unconstrained



(c)



(d)

R code: complement

Palmer & Gough

$$H_1 : \mu_1 > \mu_2 > \mu_3,$$

$$H_c : \text{not } H_1.$$

GORIC

```

H1 <- 'group1 > group2 > group3'
# vs its complement (default in case of one hypothesis)
#
# GORIC (using goric function in restriktor package)
library(restriktor)
set.seed(123) # Set seed value
goric.PandG_1c <- goric(fit.PandG,
                        hypotheses = list(H1))
    
```


Extra: complement & 'Heq' (3/4)

Palmer & Gough

R output (partly)

```
## restriktor (0.6-50): generalized order-restricted information criterion:
##
## Results:
##      model      loglik  penalty    goric  loglik.weights  penalty.weights
## 1      Heq   -193.703    3.000  393.406         0.123         0.452
## 2      H1   -191.893    3.500  390.786         0.753         0.274
## 3 complement -193.703    3.500  394.406         0.123         0.274
##      goric.weights  goric.weights_without_heq
## 1          0.188
## 2          0.698
## 3          0.114
##
## Conclusion:
## - The order-restricted hypothesis H1 is the best in the set, as it has the h
## * 'H1' is 6.109 times more supported than 'complement'.
```


Hands-on/Demo (2): GORIC - H_m vs complement

Let's practice.

- If needed: Start Rstudio again (optional: make project) and then also load packages again.
- Optional: Open 'Tutorial_GORIC_restriktor_ANOVA.html', 'PalmerAndGough_and_Lucas.R', and/or '**Hands-on_2_GORIC_Compl_ANOVA_restriktor.R**'.
- Read and inspect data.
Use Data_PalmerAndGough.txt and/or **Data_Lucas.txt**.
- Run model (lm()).
- Specify hypotheses (make up your own). E.g., for Lucas:

$$H_1 : \mu_5 = \mu_3 > \{\mu_1, \mu_4\} > \mu_2$$

Note: Use names used in the model.

- Run goric()
- Inspect and interpret output.

GORICA

GORIC: Normal linear models

GORIC can easily be applied to normal linear models (e.g., ANOVA models or regression models).

GORIC: Other statistical models

In case of other statistical models (e.g., a SEM model), more cumbersome to calculate maximized order-restricted log likelihood and thus GORIC.

GORICA: All statistical models

Therefore, GORICA: asymptotic expression for GORIC.
Can be used for all types of statistical models.

Reference:

Altınışık, Y., Van Lissa, C. J., Hoijtink, H., Oldehinkel, A. J., and Kuiper, R. M. (2021). Evaluation of inequality constrained hypotheses using a generalization of the AIC. *Psychological Methods*, 26(5), 599–621.

<https://doi.org/10.1037/met0000406>

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GORICA

Similarities with GORIC

- Form: $GORICA_m = -2 \text{ fit} + 2 \text{ complexity}$.
- Broad type of restrictions.

Differences compared to GORIC

- Uses asymptotic expression of the likelihood (is a normal):
can therefore be easily applied to all types of statistical models.
Disadvantage: might work less well in case of small samples.
- Does not need data set; mle's and their covariance matrix suffice.
- Can leave out nuisance parameters (i.e., not part of hypotheses).

Note

In case of normal linear models or, otherwise, for not too small samples:
GORICA weights = GORIC weights.

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R code: GORICA

Palmer & Gough

$$H_1 : \mu_1 > \mu_2 > \mu_3,$$

$$H_c : \text{not } H_1.$$

GORICA: type = "gorica" (not default for lm objects)

```

H1 <- 'group1 > group2 > group3'
# vs its complement (default in case of one hypothesis)
#
# GORICA (using goric function in restriktor package)
library(restriktor)
set.seed(123) # Set seed value
gorica.PandG_C <- goric(fit.PandG,
                        hypotheses = list(H1),
                        type = "gorica") # needed if lm object
    
```

GORICA

Palmer & Gough

$$H_1 : \mu_1 > \mu_2 > \mu_3,$$

$$H_c : \text{not } H_1.$$

GORIC

Model	Fit	Complexity	GORICA	GORICA weights
H1	-1.90	1.81	7.42	0.79
complement	-2.34	2.69	10.06	0.21

H_1 is $0.79 / 0.21 \approx 3.74$ times more supported than its complement, that is, any other hypothesis.

Note: GORIC(A) weights are the same (ratio may differ a bit).

GORICA

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Hands-on/Demo (3): GORICA

Let's practice.

- If needed: Start Rstudio again (optional: make project) and then also load packages again.
- Optional: Open 'Tutorial_GORIC_restriktor_ANOVA.html', 'PalmerAndGough_and_Lucas.R', and/or '**Hands-on_3_GORICA_UncAndCompl_ANOVA_restriktor.R**'.
- Read and inspect data.
Use Data_PalmerAndGough.txt and/or **Data_Lucas.txt**.
- Run model (lm()).
- Specify hypotheses (make up your own). E.g., for Lucas:

$$H_1 : \quad \mu_5 = \mu_3 > \{\mu_1, \mu_4\} > \mu_2$$

Note: Use names used in the model (or overwrite those).

- Run goric(); now, add (since lm object):

```
type = "gorica"
```
- Inspect and interpret output.

GORIC in JASP: screenshot

Palmer & Gough

Data_PalmerAndGough_JASP
(C:\WER\Dev - 503 - winter - jasp en R Lab\JASP labmeeting)

Descriptives
T-Tests
ANOVA
Mixed Models
Regression
Frequencies
Factor
Bain
SEM
R (Beta)

Order Restricted Hypotheses

Enter each restriction of one hypothesis on a new line, e.g.,
factorLow == factorMid
factorMid < factorHigh
where 'factor' is the factor (or covariate) name and 'Low'/'Mid'/'High' are the factor level names.
Click on the 'plus' icon to add more hypotheses.
Click the information icon for more examples.

Syntax settings

Set for all models

☐ Include intercept
☐ Model summary

☐ Show available coefficients
☐ Marginal means

☐ Informed hypothesis tests

Model 1
+

group1 > group2 > group3

Summary for Model 1
Marginal means for Model 1
Informed hypothesis tests for Model 1

Results

ANOVA

ANOVA - Importance

Cases	Sum of Squares	df	Mean Square	F	p
group	123.960	2	61.980	4.554	0.014
Residuals	925.418	68	13.609		

Note. Type III Sum of Squares

Order Restricted Hypotheses

Model Comparison

Model Comparison Table

Model	Log-likelihood	Penalty	GORIC	Weight	Weights ratio
Model 1	-191.893	2.814	389.415	0.789	3.731
Complement	-192.338	3.686	392.048	0.211	1.000

Note. Weights ratios indicate the relative weight for each model against the "complement" model. GORIC = Generalized Order-Restricted Information Criterion (Kuiper, Hoijtink, & Silvapulle, 2011).

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GORIC and GORICA in JASP

GORIC or GORICA in the following statistical models (as part of that model)

The screenshot displays the JASP software interface with three statistical models: ANOVA, ANCOVA, and Repeated Measures ANOVA. In each model's left-hand menu, the 'Order Restricted Hypotheses' option is circled in blue. The ANOVA and ANCOVA models also have their titles circled in blue. The Repeated Measures ANOVA model has its title circled in blue. On the right-hand side of the Repeated Measures ANOVA model, the 'Order Restricted Hypotheses' option is circled in red. The JASP logo is visible in the top right corner.

ANOVA

Order Restricted Hypotheses

Post Hoc Tests

Descriptives Plots

Residual Plots

Marginal Means

ANOVA

Cases	Sum of Squares	df	Mean Square	F	p
Note: Type III Sum of Squares					

ANCOVA

Order Restricted Hypotheses

Post Hoc Tests

Descriptives Plots

Residual Plots

Marginal Means

ANCOVA

Cases	Sum of Squares	df	Mean Square	F	p
Note: Type III Sum of Squares					

Repeated Measures ANOVA

Assumption Checks

Contrasts

Order Restricted Hypotheses

Post Hoc Tests

Descriptives Plots

Residual Plots

Marginal Means

Simple Main Effects

Repeated Measures ANOVA

Cases	Sum of Squares	df	Mean Square	F	p
Note: Type III Sum of Squares					

Between Subjects Effects

Cases	Sum of Squares	df	Mean Square	F	p
Note: Type III Sum of Squares					

Order Restricted Hypotheses

Table of Contents

Generalized Order-Restricted Information Criterion (GORIC)

GORIC weights

Examples with R code

GORICA

GORIC(A) in JASP

End & Extra

The End

GORIC(A)

Thanks for listening!

Are there any questions?

Websites

<https://github.com/rebeccakuiper/Tutorials>

www.uu.nl/staff/RMKuiper/Extra1

www.uu.nl/staff/RMKuiper/Extra2

informative-hypotheses.sites.uu.nl/software/goric/

E-mail

r.m.kuiper@uu.nl

Note on possibilities multiple studies

- Update GORIC(A) values & weights.
More data collected: (re-)calculate.
- Update hypotheses.
First data set (or a part of it) generates one or more hypotheses.
Other data set (or part) used to determine evidence / support.
Download 'Tutorial_GORIC_restriktor_UpdateHypo.html' and/or
'Hands-on_4_GORIC_UpdateHypo_restriktor.R' from
<https://github.com/rebeccakuiper/Tutorials>.
- Aggregate evidence for hypotheses.
Aggregate the support for theories (diverse designs allowed).
Bear in mind: Meta-analysis aggregates parameter estimates or
effect sizes which need to be comparable (often same designs
required).
Download 'Tutorial_GORIC_restriktor_evSyn.html' and/or
'Hands-on_4_GORIC_evSyn_restriktor.R' from
<https://github.com/rebeccakuiper/Tutorials>.

Extra material (1/2)

- GORICA on logistic Regression Model
 - Article: doi.org/10.1037/met0000406
- GORICA on SEM
 - Article: www.tandfonline.com/doi/full/10.1080/10705511.2020.1836967.
 - R scripts: github.com/rebeccakuiper/GORICA_in_SEM.
- GORICA on cross-lagged panel model (CLPM)
 - Article: doi.org/10.1111/bjep.12455.
 - R scripts: github.com/rebeccakuiper/GORICA_in_CLPM.
- GORICA on Random-Intercept CLPM (RI-CLPM)
 - Article: Sukpan, C. and Kuiper, R.M. (2023). How to evaluate causal dominance hypotheses in lagged effects models.
 - R scripts: github.com/rebeccakuiper/GORICA%20in%20RI-CLPM.
- GORICA for EffectLiteR
 - R scripts: <https://github.com/rebeccakuiper/Tutorials/tree/main/GORICA%20for%20EffectLiteR>.

Note: On github site, go to Code (green button) and download zip.

H0 vs Hm	GORIC	Weights	Example R	Failsafe	GORICA	JASP	End & Extra
○○○○○○ ○○○○○○ ○○○○○○ ○○○	○○○○○○○○○○○ ○○○○○ ○○○○○ ○○○○○ ○○○○○ ○○○	○○○ ○	○○○○○ ○○○○○ ○○○	○○○○○○○○○ ○○○○○○○○○ ○○○○○○○○○ ○○○	○○○○○○	○○○○○ ○○	○○○○○ ○○○○○○○● ○○○○○ ○○○○○○○

Note on BF as an IC

The fit in GORICA refers to the maximum log likelihood. The fit in BF is related to the maximum likelihood.

Notably, BF can be written as an IC (and vice versa):

$$\begin{aligned} -2 \log BF_{1u} &= -2 \log \frac{f_1}{c_1} \\ &= -2 \log f_1 + 2 \log c_1 \\ &= -2 \log \text{fit } H_1 + 2 \log \text{complexity } H_1. \end{aligned}$$

Note: complexity value in BF depends on prior.

1 - w_m = conditional error probability.
Like PMP, w_m depends on set of hypotheses.

Note on conditional error probability (1/3)

using PMPs and GORIC(A) weights

H_m	weights
H_1 : Sex Match	.04
H_2 : Gender Role Match	.81
H_3 : Sex Mismatch	.01
H_4 : Gender Role Mismatch	.00
H_u :	.14

The conditional error probability for preferred hypotheses H_2 is
 $1 - .81 = .19$.

Note on conditional error probability (3/3)

using PMPS and GORIC(A) weights

What does conditional error probability reflect when hypotheses overlap? Notably, when hypotheses overlap they can also share/divide support (like H_u does with any hypothesis).

My advise:

Only use error probabilities if one hypothesis versus its complement.
(Or when you are sure that there is no overlap in hypotheses;
and they preferably do cover all possibilities)

H_m	weights
H_2 : Gender Role Match	.82
H_c : complement H_2	.18

Properties of Methods: Simulation study

1. All methods (e.g., (G)ORIC and BMS) programmed in Fortran.
2. Generate data sets with know structure (conditions).
3. Apply methods to all data sets (within one condition).
4. Summarize results (per condition):
proportion of (correct) chosen hypotheses.

Reference:

Kuiper, R.M., Nederhof, T., and Klugkist, I. (2015). Properties of hypothesis testing techniques and (Bayesian) model selection for exploration-based and theory-based (order-restricted) hypotheses. *British Journal of Mathematical and Statistical Psychology*, 68(2), 220 – 245.

Population Values in Simulated Data Sets when $k = 3$

Values of Population Means (μ_i) and Standard Deviations (σ_i) Used to Simulate 1000 Data Sets

$k = 3$	Case Number ('ES')	μ_1	μ_2	μ_3	ES (Cohen)
	0	0.00	0.00	0.00	
	1	-0.10	0.00	0.10	.1
	2	-0.20	0.00	0.20	.2
	3	-0.30	0.00	0.30	.3
	4	-0.40	0.00	0.40	.4
	5	-0.50	0.00	0.50	.5
		σ_1	σ_2	σ_3	
	0 - 5	1.00	1.00	1.00	

NB in Case 0 $H_{30C} : \mu_1 = \mu_2 = \mu_3$ is true
in Cases 1 to 5 $H_{31C} : \mu_1 < \mu_2 < \mu_3$ is true.

Results Confirmation ($k = 3$ and $n = 50$)

$$H_{30C} : \mu_1 = \mu_2 = \mu_3$$

$$H_{31C} : \mu_1 < \mu_2 < \mu_3$$

$$H_{32C} : \mu_1 = \mu_2 < \mu_3$$

$$H_{33C} : \mu_1 < \mu_2 > \mu_3$$

$$H_{34C} : \mu_1, \mu_2, \mu_3$$

1 – Type I error (here):

preferring the true null hypothesis,
given the set of hypotheses (!).

'ES'	Method	'Prior'	H_{30C}	H_{31C}	H_{32C}	H_{33C}	H_{34C}
0	ORIC		0.662	0.065	0.119	0.120	0.034
0	BMS	3	0.865	0.012	0.075	0.036	0.012
0	BMS	2	0.774	0.022	0.124	0.058	0.022
0	BMS	1	0.656	0.047	0.147	0.120	0.033
2	ORIC		0.148	0.506	0.311	0.029	0.006
2	BMS	3	0.299	0.299	0.371	0.031	0.001
2	BMS	2	0.252	0.328	0.381	0.040	0.000
2	BMS	1	0.125	0.494	0.321	0.063	0.000
5	ORIC		0.000	0.955	0.044	0.001	0.000
5	BMS	3	0.000	0.887	0.112	0.001	0.000
5	BMS	2	0.001	0.890	0.107	0.002	0.000
5	BMS	1	0.000	0.928	0.070	0.002	0.000

Results Confirmation ($k = 3$ and $n = 50$)

$$H_{30C} : \mu_1 = \mu_2 = \mu_3$$

$$H_{31C} : \mu_1 < \mu_2 < \mu_3$$

$$H_{32C} : \mu_1 = \mu_2 < \mu_3$$

$$H_{33C} : \mu_1 < \mu_2 > \mu_3$$

$$H_{34C} : \mu_1, \mu_2, \mu_3$$

1 – Type II error (here):
 preferring the true hypothesis,
 given the set of hypotheses (!).

'ES'	Method	'Prior'	H_{30C}	H_{31C}	H_{32C}	H_{33C}	H_{34C}
0	ORIC		0.662	0.065	0.119	0.120	0.034
0	BMS	3	0.865	0.012	0.075	0.036	0.012
0	BMS	2	0.774	0.022	0.124	0.058	0.022
0	BMS	1	0.656	0.047	0.147	0.120	0.033
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2	BMS	3	0.299	0.299	0.371	0.031	0.001
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2	BMS	1	0.125	0.494	0.321	0.063	0.000
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